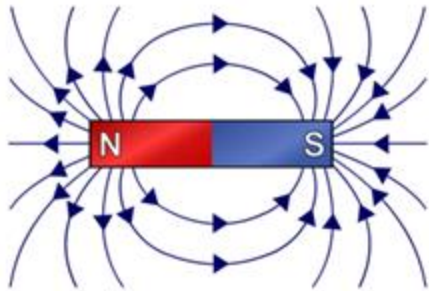
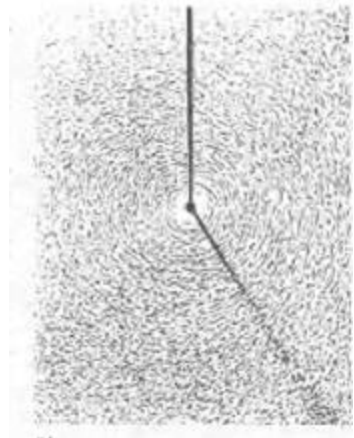
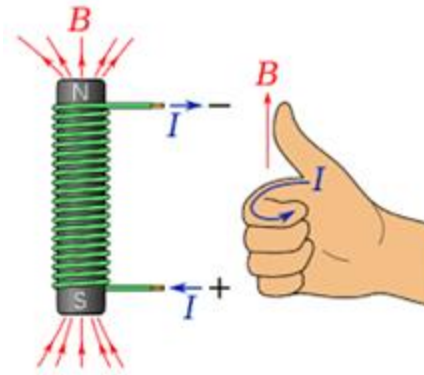
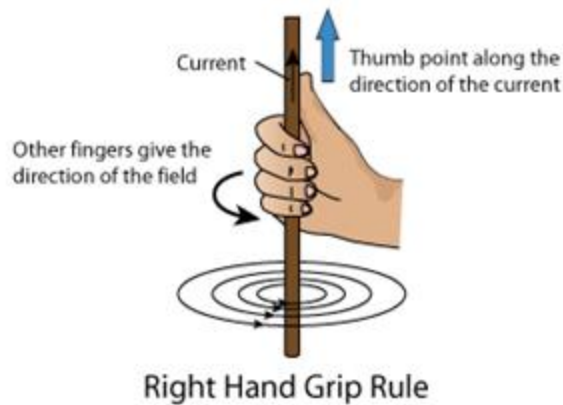
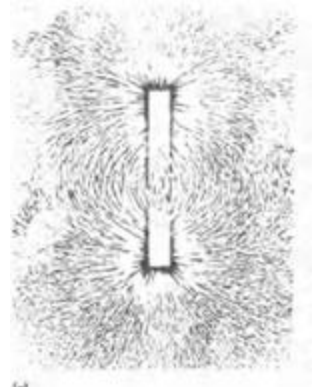


Magnetic field lines

The *magnetic field* is an abstract entity that describes the influence of magnetic forces in a region. *Magnetic field lines* are a visual tool used to represent magnetic fields. They describe the direction of the magnetic force on a north monopole at any given position.

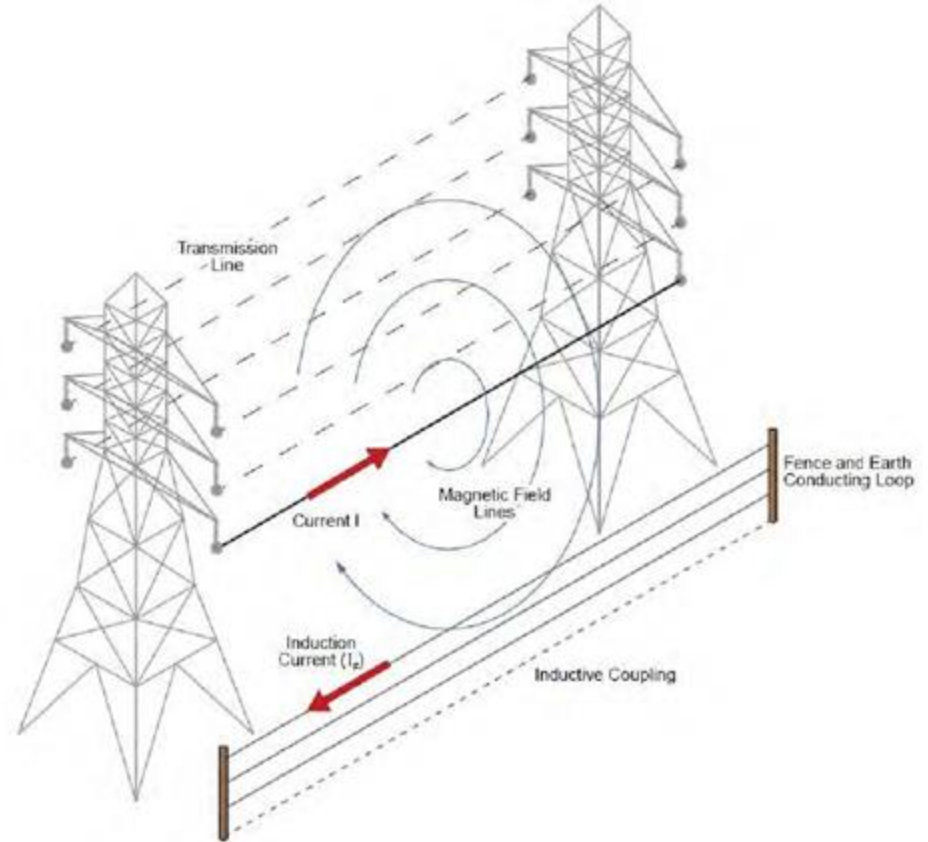
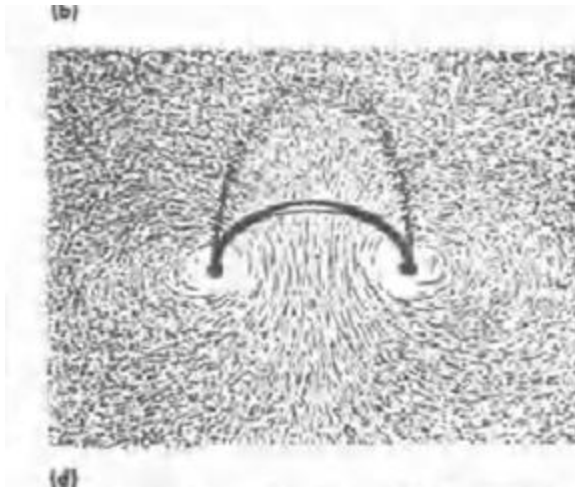


Outside a bar magnet, the lines of force leave the north pole and return at the south pole. (Inside the magnet, the situation is more complicated)



From: *Introduction to Magnetism and Magnetic Materials* – By David Jiles
<https://brilliant.org/wiki/magnetic-field-lines/>

Magnetic field lines



Simplified version for illustration and not drawn to scale

From: *Introduction to Magnetism and Magnetic Materials* – By David Jiles
<https://www.nhsec.nh.gov/>

Origin of magnetism

1.1.1 Generation of a magnetic field

What causes magnetic fields in the first place?

A magnetic field is produced whenever there is electrical charge in motion. This can be due to an electrical current flowing in a conductor for example, as was first discovered by Oersted in 1819 [1].

What about the permanent magnets??

A magnetic field is also produced by a permanent magnet. In this case there is no conventional electric current, but there are the orbital motions and spins of electrons (the so called 'Amperian currents') within the permanent magnet material which lead to a magnetization within the material and a magnetic field outside.

We will have to delve through the atoms!

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The Stern-Gerlach experiment

Stern and Gerlach: How a Bad Cigar Helped Reorient Atomic Physics

The history of the Stern–Gerlach experiment reveals how persistence, accident, and luck can sometimes combine in just the right ways.

Bretislav Friedrich and Dudley Herschbach

tistical physics. However, Stern was shocked by the iconoclastic atomic model of Niels Bohr. Shortly after it appeared in mid-1913, Stern and his colleague Max von Laue made an earnest vow: "If this nonsense of Bohr should in the end prove to be right, we will quit physics!"⁴ When Ein-

Bohr atomic model Quantized energy levels

Stern wanted to prove Bohr wrong and conducted an experiment with Gerlach (they worked at Max Born Institute)

periments to test various geometrical configurations. Born doubted that the deflection experiment would prove worthwhile. Gerlach's response was to quote a favorite saying, later apt for the SGE as well: "No experiment is so dumb, that it should not be tried."⁵

The Stern-Gerlach experiment

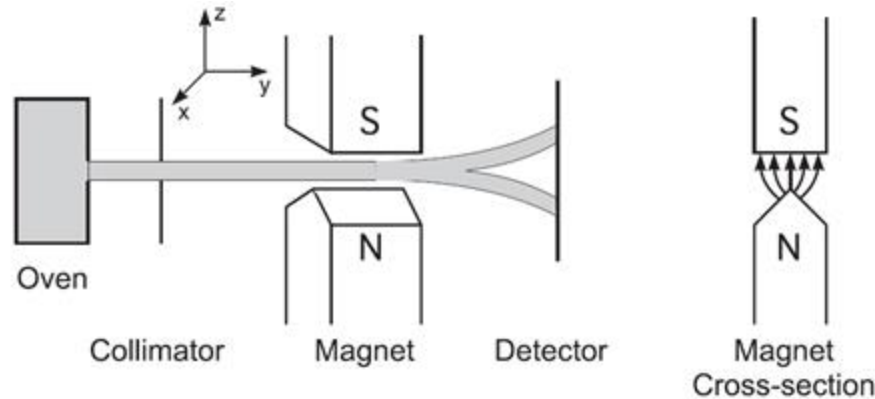


Figure 1.1 Stern-Gerlach experiment to measure the spin component of neutral particles along the z-axis. The magnetic cross-section at right shows the inhomogeneous field used in the experiment.

1.1 Stern-Gerlach experiment

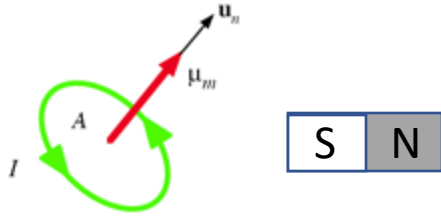
In 1922 Otto Stern and Walther Gerlach performed a seminal experiment in the history of quantum mechanics. In its simplest form, the experiment consists of an oven that produces a beam of neutral atoms, a region of space with an inhomogeneous magnetic field, and a detector for the atoms, as depicted in Fig. 1.1. Stern and Gerlach used a beam of silver atoms and found that the beam was split into two in its passage through the magnetic field. One beam was deflected upwards and one downwards in relation to the direction of the magnetic field gradient.

To understand why this result is so at odds with our classical expectations, we must first analyze the experiment classically. The results of the experiment suggest an interaction between a neutral particle and a magnetic field. We expect such an interaction if the particle possesses a magnetic moment μ . The potential energy of this interaction is $E = -\mu \cdot \mathbf{B}$, which results in a force $\mathbf{F} = \nabla(\mu \cdot \mathbf{B})$. In the Stern-Gerlach experiment, the magnetic field gradient is primarily in the z-direction, and

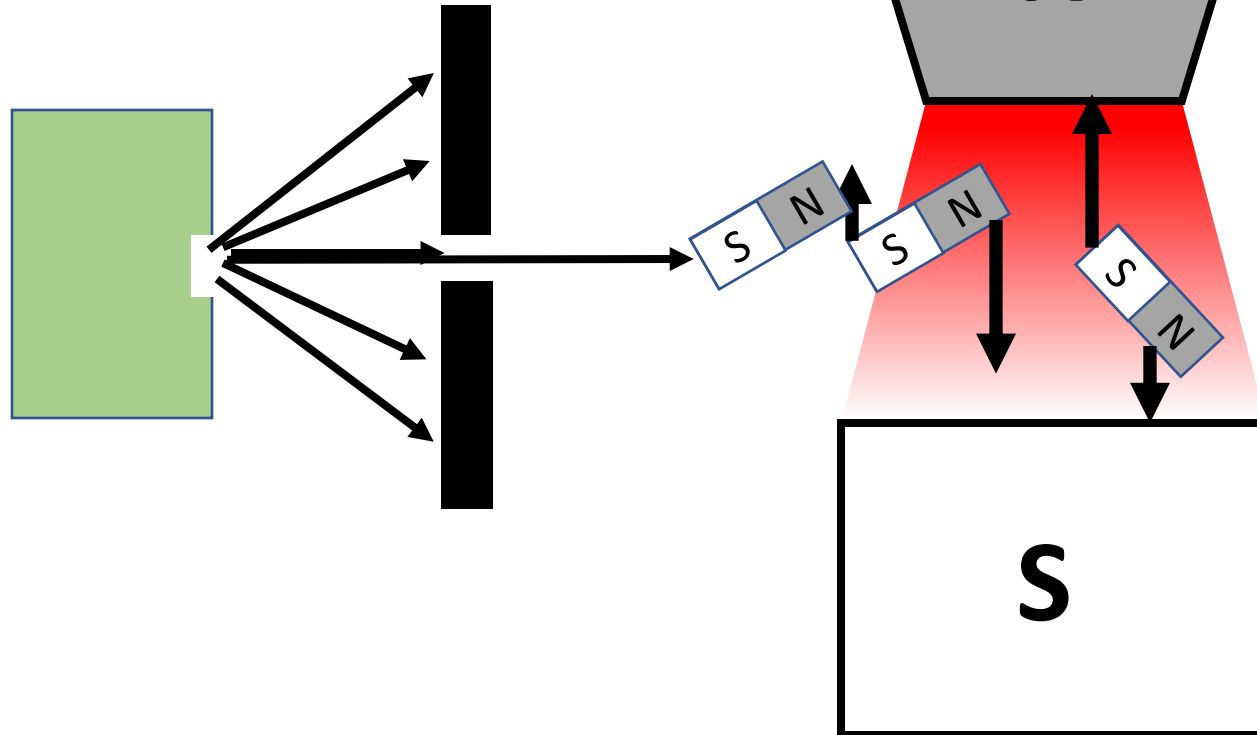
- Inhomogeneous magnetic field
- Ag atoms were used in the experiment (produced by effusion of metallic vapor from an oven heated to 1000°C)
- collimated by two narrow slits (0.03 mm wide) and traversed a deflecting magnet 3.5 cm long with field strength about 0.1 tesla and gradient 10 tesla/cm

$$F_z = \frac{\partial}{\partial z}(\mu \cdot \mathbf{B})$$
$$\cong \mu_z \frac{\partial B_z}{\partial z}$$

The Stern-Gerlach experiment



Each atom can be considered as a bar magnet



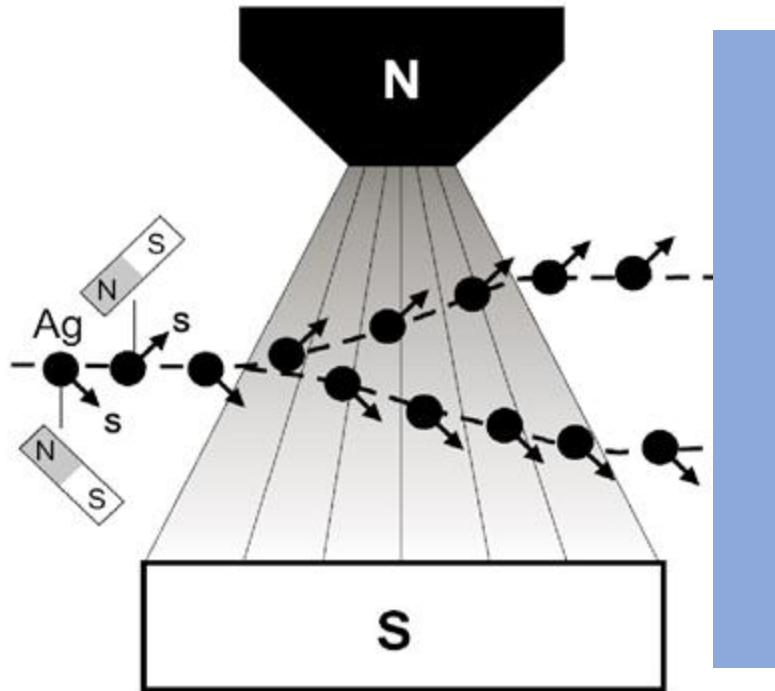
The Stern-Gerlach experiment

**THE SPIN,
A QUANTUM MAGNET**

All the animations and explanations on
www.toutestquantique.fr

**Expectation and actual result!!
Doesn't match**

The Stern-Gerlach Experiment



A stream of Ag atoms
passing through a
nonuniform magnetic field
splits into two

This experimental finding has
significant contribution
towards the discovery of
electron spin.

Electron spin cannot be measured directly. From the spin magnetic moment measurement (Stern-Gerlach apparatus), it is concluded whether the electron spin points up/down.

The Stern-Gerlach Experiment



Stern-Gerlach memorial plaque at the University of Frankfurt. The drawing shows the original Stern-Gerlach experiment in which the Ag atom beam is passed along the long- length of the external magnet to increase the time spent in the nonuniform field, and hence increase the splitting. The photo on the lower right is Otto Stern (1888-1969), standing and enjoying a cigar while carrying out an experiment. Otto Stern won the Nobel prize in 1943 for development of the molecular beam technique



Courtesy of the Niels Bohr Archive
Actual experimental result recorded on a photographic plate by Stern and Gerlach.

Plaque photo courtesy of Horst Schmidt-Böcking from B. Friedrich and D. Herschbach, "Stern and Gerlach: How a Bad Cigar Helped Reorient Atomic Physics", *Physics Today*, December 2003, p.53-59.

From *Principles of Electronic Materials and Devices, Fourth Edition*, S.O. Kasap (© McGraw-Hill Education, 2018)

Where does the Cigar comes from!!!



Courtesy of AIP Emilio Segrè Visual Archives, Segrè Collection

The late Edwin Land, when told the cigar story many years ago, immediately responded: “I don’t believe it!”

“I was then the equivalent of an assistant professor. My salary was too low to afford good cigars, so I smoked bad cigars. **These had a lot of sulfur in them, so my breath on the plate turned the silver into silver sulfide, which is jet black, so easily visible. It was like developing a photographic film.**”

Figure 5. Reenactment of the Stern–Gerlach cigar episode by the authors. Bretislav Friedrich holds the slide as Dudley Herschbach blows sulfurous cigar breath onto a silver-coated glass slide to test his hearing (or Otto Stern’s telling) of the story more than 40 years ago. The silver film turns out to require exposure to cigar smoke (not simply sulfurous breath) to form any visible contrast between the masked (light) part of the slide—shaped in the form of the magnet pole pieces—and the outer (dark) part of the slide exposed to the smoke (see inset). (Courtesy of Doo Soo Chung and Sunil Sheth.)



From *Principles of Electronic Materials and Devices, Fourth Edition*, S.O. Kasap (© McGraw-Hill Education, 2018) and Friedrich, Bretislav, and Dudley Herschbach. “Stern and Gerlach: How a Bad Cigar Helped Reorient Atomic Physics.” *Physics Today*, vol. 56, no. 12, 2003, pp. 53–59., doi:10.1063/1.1650229.

The Stern-Gerlach Experiment

Reactions to the Stern–Gerlach Experiment

The following quotes from James Franck, Niels Bohr, and Wolfgang Pauli are among the messages that Walther Gerlach received in immediate response to postcards (like the one shown in figure 4) he had sent;¹⁰ the quote from Arnold Sommerfeld appeared in the 1922 edition of his classic book;¹⁷ that from Albert Einstein is in a March 1922 letter to Born;¹⁸ that from I. I. Rabi is from reference 8, page 119. (See also Rabi's obituary for Otto Stern in *PHYSICS TODAY*, October 1969, page 103.)

Through their clever experimental arrangement Stern and Gerlach not only demonstrated *ad oculos* [for the eyes] the space quantization of atoms in a magnetic field, but they also proved the quantum origin of electricity and its connection with atomic structure.

—Arnold Sommerfeld (1868–1951)

The most interesting achievement at this point is the experiment of Stern and Gerlach. The alignment of the atoms without collisions via radiative [exchange] is not comprehensible based on the current [theoretical] methods; it should take more than 100 years for the atoms to align. I have done a little calculation about this with [Paul] Ehrenfest. [Heinrich] Rubens considers the experimental result to be absolutely certain.

—Albert Einstein (1879–1955)

More important is whether this proves the existence of space quantization. Please add a few words of explanation to your puzzle, such as what's really going on.

—James Franck (1882–1951)

I would be very grateful if you or Stern could let me know, in a few lines, whether you interpret your experimental results in this way that the atoms are oriented only parallel or opposed, but not normal to the field, as one could provide theoretical reasons for the latter assertion.

—Niels Bohr (1885–1962)

This should convert even the nonbeliever Stern.

—Wolfgang Pauli (1900–58)

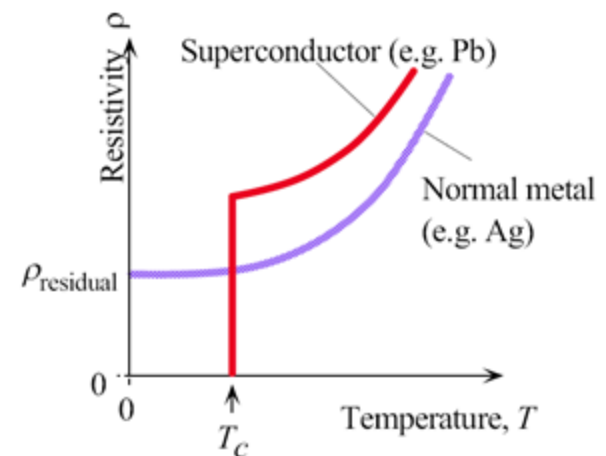
As a beginning graduate student back in 1923, I . . . hoped with ingenuity and inventiveness I could find ways to fit the atomic phenomena into some kind of mechanical system. . . . My hope to [do that] died when I read about the Stern–Gerlach experiment. . . . The results were astounding, although they were hinted at by quantum theory. . . . This convinced me once and for all that an ingenious classical mechanism was out and that we had to face the fact that the quantum phenomena required a completely new orientation.

—Isidor I. Rabi (1898–1988)

Superconductivity

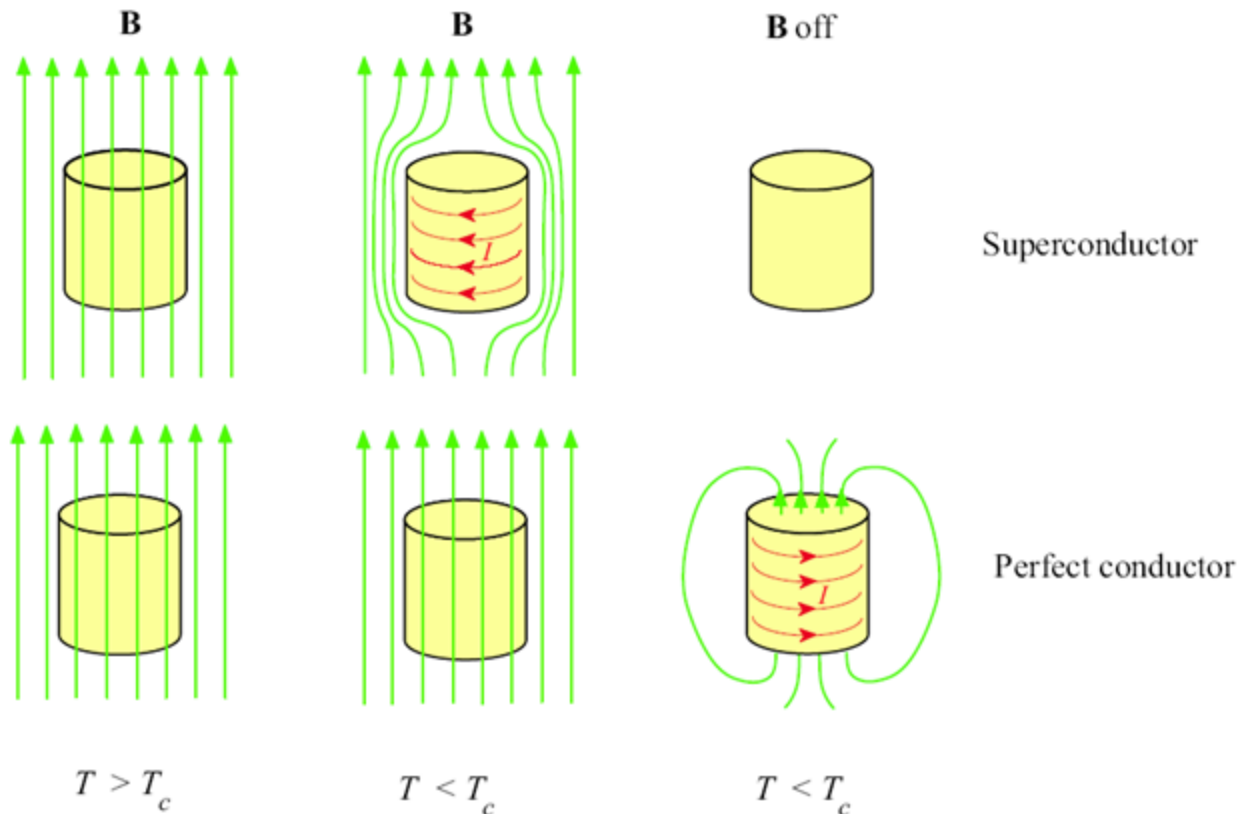
In 1911 Kamerlingh Onnes at the University of Leiden in Holland observed that when a sample of mercury is cooled to below 4.2 K, its resistivity totally vanishes and the material behaves as a **superconductor**, exhibiting no resistance to current flow. Other experiments since then have shown that there are many such substances, not simply metals, that exhibit superconductivity when cooled below a **critical temperature** T_c that depends on the material. On the other hand, there are also many conductors, including some with the highest conductivities such as silver, gold, and copper, that do not exhibit superconductivity. The resistivity of these **normal conductors** at low temperatures is limited by scattering from impurities and crystal defects and saturates at a finite value determined by the residual resistivity. The two distinctly different types of behavior are depicted in Figure 8.56. Between 1911 and 1986, many different metals and metal alloys had been studied, and the highest recorded critical temperature was about 23 K in a niobium–germanium compound

Okay but such low temperature!



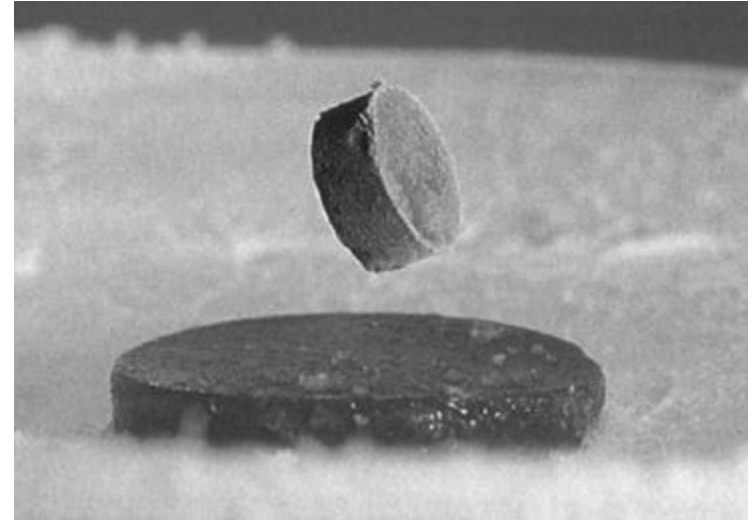
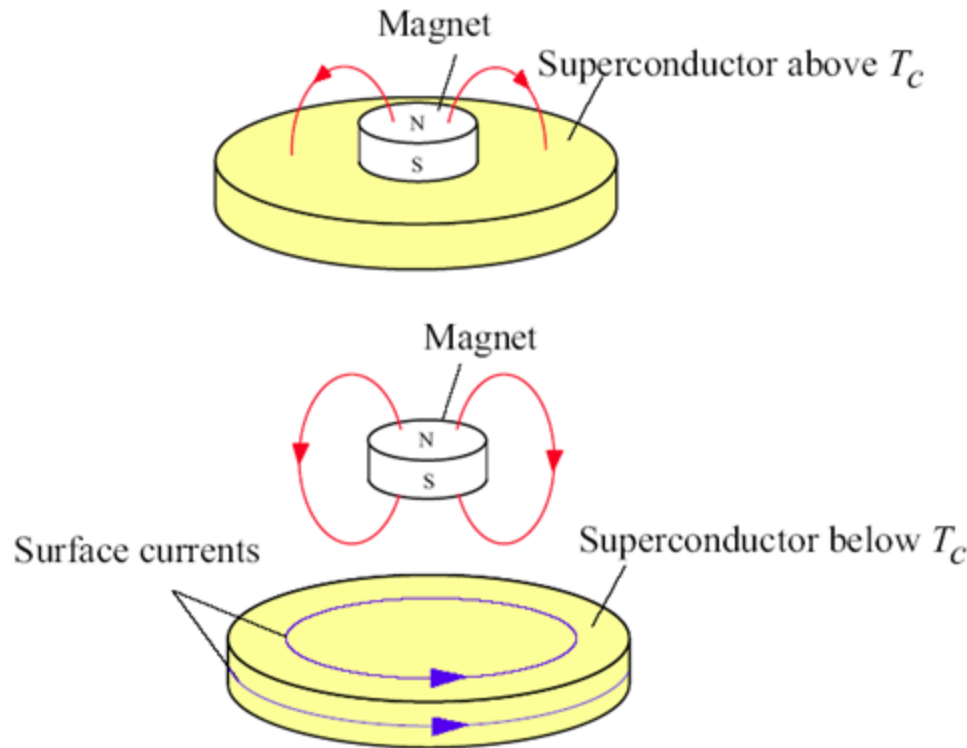
Meissner effect

The vanishing of resistivity is not the only characteristic of a superconductor. A superconductor cannot be viewed simply as a substance that has infinite conductivity below its critical temperature. A superconductor below its critical temperature expels all the magnetic field from the bulk of the sample as if it were a perfectly diamagnetic substance. This phenomenon is known as the **Meissner effect**. Suppose



The Meissner effect. A superconductor cooled below its critical temperature expels all magnetic field lines from the bulk by setting up a surface current. A perfect conductor ($\sigma = \infty$) shows no Meissner effect.

Meissner Effect and Magnetic Levitation



Left: A magnet over a superconductor becomes levitated. The superconductor is a perfect diamagnet which means that there can be no magnetic field inside the superconductor.

Right: Photograph of a magnet levitating above a superconductor immersed in liquid nitrogen (77 K). This is the Meissner effect.

Photo courtesy of Professor Paul C. W. Chu, University of Houston